

QUANTUM ENCRYPTION MODEL

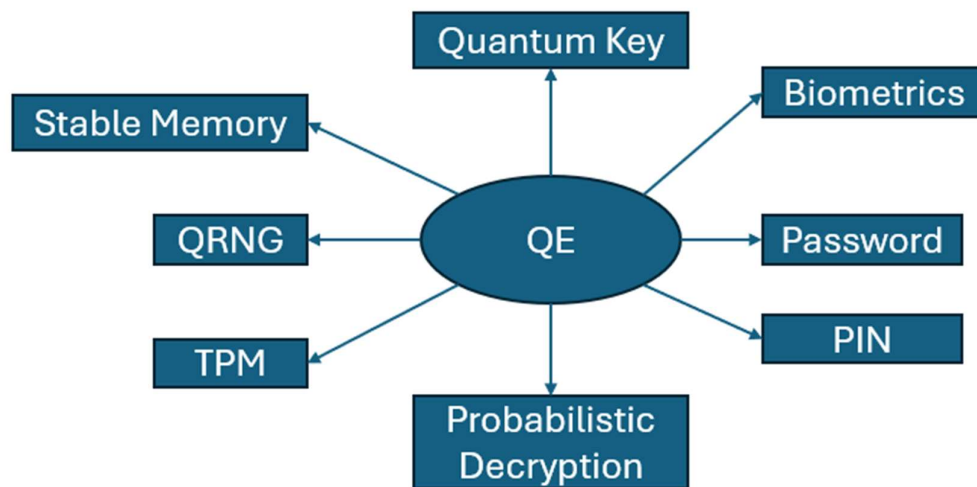
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Introduction

This quantum encryption model otherwise known as QE is designed to secure data on a system. This quantum encryption model was developed currently quantum computing can be used to crack multiple classical encryption algorithms. This means that quantum encryption should be developed to counter the offensive capabilities of computing. This quantum encryption mode does not change using classical authentication methods or using QKD as a secure key distribution method. This quantum encryption model will focus on the phases of encryption, decryption and quantum key management.

This quantum encryption model cannot promise perfect security or reliability as no security tool or feature can do so. However, this quantum encryption can promise to try create a quantum encryption system. This model was created for high-risk systems such as government-level data. The technical specifications and explanations in this documentation are made and targeted towards quantum engineers and experts. This quantum encryption model can be adapted based on the implementation and real-life conditions.



This spider diagram shows the main concepts involved in the quantum encryption model. For those not familiar with quantum computing, QRNG stands for quantum random number generator. There will be more information on the quantum encryption model in the sections below.

Quantum Encryption Model

- This quantum encryption model uses a hybrid of quantum and classical computing concepts.
- This model has 2 phases
- The first phase is encryption.
- First, the quantum processor fetches the plaintext data from the correct category in memory
- Second, the quantum processor uses quantum random number generators to generate a truly random number
- Third, the quantum processor uses this number to shift the letters to create a completely random pattern.
- Fourth, the quantum processor creates a quantum key by using a mixture of superposition, entanglement and measure.
- Qubits are superposed and entangled using the quantum processing cycle and specific gates.
- After the qubits are ready, they are measured creating a random bit string of 0s and 1s which include lengths such as 32-bit, 128-bit or 256-bit.
- Fifth, the quantum processor works with classical computing components to tie the key to an authentication method.
- For example, authentication methods can include passwords, PINs, MFA, Biometrics. You may also include rate-limiting.
- Sixth, the quantum processor uses the quantum key to encrypt the data to make it inaccessible to anyone trying to access the data.
- Seventh, the quantum key is stored within TPM in 0s and 1s for easier storage allowing the quantum key to be stored safely.
- This ensures that the key is secured and stored correctly and reliably.
- The second phase is decryption.
- First, the quantum processor fetches the encrypted data from storage or memory as needed.
- Second, the user must authenticate using their password, PIN or biometrics although this varies throughout the system.
- Third, after the user authenticates the quantum key is fetched from TPM where it was secured.
- Fourth, the processor uses the quantum key and runs multiple decryption attempts.
- Multiple decryption attempts are used as the pattern is random and will require multiple attempts to get the correct output.
- Fifth, the processor decrypts the data successfully and displays it to the user for access and usage.
- This decryption process is slow as it is probabilistic however over time as quantum processing gets faster more reliable it may speed up.

Quantum Encryption Model Trade-offs

- This quantum encryption model has trade-offs just like any other system I have created.
- There may further implementation challenges as quantum computing is still in early stages of development.
- I have identified 5 trade-offs.
- The first trade-off is cost.
- This trade-off exists because creating hybrid quantum and classical computing systems is extremely expensive.
- This is because materials for classical hardware are already expensive while quantum computing components and cooling systems are extremely expensive.
- This trade-off can be managed by allocating a budget, using pre-existing systems and be as efficient as possible within the system.
- The second trade-off is time-consumption.
- This trade-off exists because creating hybrid quantum and classical computing systems on this level will take years.
- This is because classical processing and authentication must be combined with quantum processing and security.
- This trade-off can be managed by using project management, multiple teams of people and avoiding complexity where possible.
- The third trade-off is complexity
- This trade-off exists as combining classical hardware and quantum computing systems is difficult and confusing to figure out.
- This is because classical computing and quantum computing are fundamentally different and combining them requires knowledge in both systems.
- This trade-off can be managed by allocating teams, ensuring that both systems are understood and using structured project management processes.
- The fourth trade-off is compatibility.
- This trade-off exists because this system cannot be used on a purely classical or quantum system.
- This is because certain components from each type of computing are needed for this system to function correctly and efficiently.
- This trade-off can be managed by accepting that this system won't be pure and by using structured and targeted deployment.
- The fifth trade-off is decryption speed.
- This trade-off exists because this system uses probabilistic decryption which is slower than normal decryption.
- This is because instead of there being a predictable pattern quantum encryption uses random patterns meaning it takes more effort to decrypt.
- This trade-off can be managed varying key length depending on the system, increasing quantum processing speeds and accepting it won't be as fast.